

General Championship – Data Analytics (2026)

Industry Case Partner: Frammer AI

Theme: Building a Product Usage Analytics Dashboard for Media Operations

1) Background & Context

Frammer AI is a B2B platform used by media teams to convert long-form and live content into short-form vertical videos, chapters, summaries, and other publish-ready outputs with metadata — with just one click.

Frammer already has an internal analytics dashboard used to track:

- **Uploaded / Processed / Published videos** (count and hours)
- **Duration-based usage**
- **Channel-wise performance**
- **Video category/type breakdowns** (e.g., reels, shorts, chapters, summaries, viral videos)
- **Video-level details** (headline, source, team, uploader, publish status, platform, etc.)

As the number of clients, channels/workspaces, users, languages, and output types grows, Frammer now wants a more scalable, user-friendly, and insight-ready dashboard that is easier to navigate and can support future feature additions.

2) Problem Statement

You are the analytics product team for Frammer AI. Your task is to design and build an improved analytics dashboard solution (**data model + dashboard + insights**) that:

- Improves the current reporting experience for internal and client-facing use
- Supports **multi-dimensional drill-downs** and clear KPI tracking
- Is **easy to navigate**, even for non-technical users
- Has a **clean, modular structure** so new metrics/dimensions can be added later
- Surfaces **actionable insights**, not just raw counts and charts

The solution should be designed as if it will be used by:

- Founders / Leadership
- Client Success teams
- Operations teams
- Editorial / Production teams
- Tech / Product teams

Important context: A single client may have **multiple channels/workspaces**, and each channel/workspace may have **different users, languages, and usage patterns**. The dashboard should handle this structure clearly.

The solution must follow **analytics best practices** and be built in a **modular, scalable** manner, so that new metrics, features, dimensions, and reporting layers can be added easily in the future without redesigning the dashboard.

3) Current Dashboard Context (Reference for Improvement)

Frammer's current dashboard includes elements such as:

A) Filters & Navigation

- Date Range
- Comparison Date Range
- Company (Client)
- Channel
- Users
- Language
- Video Type
- Dimension / Dimension Filter
- Published Filter

B) Summary KPI Blocks

- Uploaded Videos (count + duration + published subset)
- Processed Videos (count + duration + published subset)
- Published Videos (count + duration)

C) Output Type / Category Cards

Examples in current UI:

- Reels
- Chapters
- MKM Videos
- Summary Videos
- Shorts Videos
- Viral Videos
- Replaced Videos
- Non-Billable Videos

Each card currently shows:

- Total count
- Duration
- Published count (and published duration)

D) Trend & Breakdown Views

- Day-wise video count chart
- Channel-wise uploaded / processed / published comparison chart
- Channel-wise summary table

E) Video-Level Detail Table

Current table includes columns such as:

- Headline
 - Source
 - Published (Yes/No)
 - Team Name
 - Type
 - Uploaded By
 - Video ID
 - Published Platform
 - Published URL
- (With pagination and export options)

4) Core Challenge for Teams

The challenge is not just to recreate the current dashboard, but to improve it in a meaningful way. Teams must propose and build a dashboard that is:

1) More Intuitive

- Better information hierarchy
- Cleaner navigation from overview → detail
- Better grouping of filters and metrics

2) More Analytical

- Stronger comparisons (time-period comparisons, trends, growth/decline)
- Better breakdowns across client / channel / user / language / type / platform
- More insight-oriented visuals and summaries

3) More Scalable

- Easy to add new output types, dimensions, clients, and metrics
- Clear metric definitions and reusable logic
- Data model that avoids duplication/confusion

4) More Operationally Useful

Should help identify:

- Drop-offs (Processed vs Published)
- Underused channels/users
- High-volume / low-publish patterns
- Data quality gaps (e.g., Unknown team names, missing fields)
- Videos with errors and type of errors (if error data is available)

5) More Accessible (Natural Language / Prompting + Vector Search)

The dashboard should ideally support **natural language / conversational queries (prompt-based)** so non-technical users can ask questions in plain English. Teams should implement this using **vector search / semantic retrieval** wherever applicable.

Examples of queries:

- “Show channel-wise usage hours for interviews vs speeches last month”
- “Which channels have the biggest processed vs published gap?”

- “Top 10 videos by duration and their publish status”
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5) Multi-Dimension Requirement (Important)

A key requirement is multi-dimensional analysis, including **double-dimension breakdowns**.

Teams should design the dashboard so users can analyze usage by:

- **Dimension 1** (example: Channel)
- **Dimension 2** (example: Input Type / content category such as Speech, Interview, Special Report, etc.)

Example use case

A user selects:

- Dimension 1 = Channel
- Dimension 2 = Input Type

The dashboard should then show the **channel-wise breakup by input type**, for both:

- Video counts
- Usage hours/duration
(and optionally publish conversion %)

This should work similarly for other combinations as well, for example:

- Channel × Language
- User × Output Type
- Team × Platform
- Channel × Published Status

The dashboard should support this in a way that is easy to use and easy to extend later.

6) Objectives (What the dashboard should help answer)

A) Usage & Adoption

- How many videos are being uploaded, processed, and published over time?
- What is total duration processed (daily/weekly/monthly)?
- Which clients / channels / users are driving usage?
- How does usage compare vs previous time periods?

B) Output Mix & Content Type Trends

- What is the split by output type (reels, shorts, summaries, chapters, etc.)?
- What is the split by input type (speech, interview, special report, etc.)?
- Which output/input categories are growing or declining?
- Which formats are used most by which channels/users?

C) Publishing Funnel & Efficiency

- What is the gap between processed and published content?
- Which channels or teams process high volume but publish low?
- What is the publish conversion % by channel / type / user?

D) Team / User / Language / Platform Insights

- Which users or teams contribute most volume?
- Which languages or platforms are most active?
- Are there underperforming channel-language or user-channel combinations?

E) Data Quality & Governance

- How much data is missing or marked as “Unknown”?
- Which fields frequently have missing platform / URL / team values?
- How can data quality be monitored over time?

7) Expected Deliverables

1. Working Dashboard (Mandatory)

Preferably **3–5 pages/tabs** (or equivalent navigation), for example:

1. **Executive Summary**
 - Top KPIs, trends, comparisons, alerts
2. **Usage & Trends**
 - Time-based charts (count + duration), period comparison
3. **Client / Channel / User Analysis**
 - Breakdown and drill-down views
4. **Type Mix & Publishing Funnel**
 - Processed vs Published, conversion, output type + input type mix
5. **Video-Level Detail / Explorer**
 - Searchable, filterable detail table with export-ready logic

Additionally, teams should provide:

- A **natural language query / conversational interface (web-based)**, where users can type prompts/questions and receive relevant results (charts/tables/answers) using **vector search / semantic retrieval** wherever applicable.
- The output should clearly show what filters/dimensions were applied.

2. Data Model + Metric Documentation

- Metric dictionary (definitions + formulas)
- Dimension dictionary
- Data model/schema (recommended: star schema)
- Assumptions clearly documented

3. Insights Presentation (8–12 slides)

- Dashboard walkthrough
- Top insights
- Improvement recommendations
- “Next layer” roadmap

4. Code / Logic / Build Notes (if applicable)

- SQL / Python / ETL logic / dashboard build notes
- Version-controlled repo preferred
- No public upload of sensitive raw data

8) Dashboard Design & Analytics Best Practices (Important)

Teams should follow strong analytics and product-thinking principles.

A) UX / Usability

- Clear visual hierarchy (overview first, details later)
- Consistent labels and naming
- Minimal clutter
- Sensible defaults for filters
- Easy export / sharing workflows

B) Metric Clarity

- One clear definition per KPI
- Avoid double-counting
- Count and duration metrics should be clearly separated
- Use percentage metrics where useful (e.g., publish rate)

C) Extensibility (Must-have)

The solution should be modular and support future additions with minimal rework, such as:

- New output types
- New input categories
- New dimensions (region, client segment, plan type, etc.)
- SLA / performance metrics
- Error/failure/retry metrics
- Billing / billable vs non-billable analytics
- Additional reporting layers/views

D) Data Quality Layer

Strong submissions should include a basic data quality monitoring approach:

- Missing values by field
- “Unknown” buckets
- Duplicate video/job IDs
- Invalid/malformed publish URLs
- Null platform/team mappings

E) Natural Language Analytics (NLQ) & Vector Search (Must-have / Bonus-worthy)

- The solution should support a **prompt-based querying layer** so stakeholders can ask questions in plain English.
 - Teams may implement this using **vector search / semantic search** (with appropriate safeguards), for example: retrieval over metric definitions/help text, mapping user queries to correct filters/dimensions, and returning relevant tables/charts/insights.
 - The natural language layer should be **usable, accurate, and explainable** (i.e., show what filters/dimensions were applied).
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9) Data & Access (What Frammer will provide)

Frammer AI will provide an anonymized dataset and/or read-only extracts for competition use.

Indicative fields (based on current dashboard)

- video_id / job_id
- headline
- source / source_url
- processed_at / uploaded_at / published_at (as available)
- published_flag
- company / client
- channel / workspace
- user / uploaded_by
- team_name
- language
- input_type (e.g., speech, interview, special report, etc.)
- output_type (e.g., reels, shorts, chapters, summaries, etc.)
- duration
- published_platform
- published_url
- billable_flag / non_billable_flag (if available)

Teams may also be given:

- Reference mapping tables (channel → client, user → team, type categories)
- Data dictionary (if available)

Data handling rules

- Data will be anonymized and shared only for case execution
 - Teams must treat data as confidential
 - No external redistribution
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10) Judging Criteria (Total: 100 Marks)

Please ensure judges' names are not included in the document.

Suggested Marking Scheme

1. **Business understanding & KPI design** – 20
(Relevance of metrics, completeness, logic)
 2. **Dashboard UX & navigability** – 20
(Ease of use, flow, structure, clarity)
 3. **Analytical depth & insight quality** – 20
(Comparisons, trends, funnel analysis, actionability, multi-dimension usage, and ability to support natural language / conversational querying where applicable)
 4. **Data quality checks & correctness** – 15
(Validation, cleaning, consistency, assumptions)
 5. **Scalability / Extensibility of architecture** – 15
(Easy addition of new KPIs, dimensions, filters, views, and reporting layers with minimal rework)
 6. **Presentation & communication** – 10
(Clarity, storytelling, executive readiness)
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11) Submission Guidelines

Teams should submit:

- **Browser-based dashboard (mandatory):**
A web-accessible output that can be opened in a standard web browser (e.g., hosted web app, web dashboard link, or browser-rendered dashboard).
It should be usable without requiring desktop software installation.
 - **Insights deck:** PDF/PPT (8–12 slides)
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- **Documentation:**
Metric dictionary + data model documentation (including assumptions)
- **Code / logic / build notes (if used):**
GitHub repo or zipped code package with clear run instructions and dependencies (including any SQL / Python / ETL / embedding + vector search components)